ABC implies an effective bound for Roth's constant for 3rd roots

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Abstract

Enrico Bombieri proved that the ABC Conjecture implies Roth's theorem in 1994. In making use of Bombieri's and Van der Poorten's explicit formula for the coefficients of the regular continued fractions of 3rd roots, we give an effective bound for Roth's constant using ABC. Roth's original proof does only prove the existence of such a constant.

1 Introduction

The proof of Roth's theorem is very long and not constructive (see the famous paper [Roth(1955)]). So there is no explicit formula for Roth's constant. It is was proved by Bombieri that Roth's theorem follows from ABC (see [Bombieri(1994)]). We give an effective bound for 3rd roots with the help of a formula by Bombieri and Van der Poorten (see [Bombieri & Van der Poorten(1975)]) for the coeffcients of the regular continued fractions of 3rd roots. And we give the relation between the epsilons of ABC and Roth's theorem. In our first paper together with Sibbertsen (see [Sibbertsen(2022)]) we have shown that a weakened form of ABC follows from Roth's theorem. This paper uses basically the same techniques and terms. The coefficients of the regular continued fraction are called b_n and the approximants $\frac{p_n}{q_n}$.

2 Roth's theorem

Roth's Theorem: Let a be an algebraic number. Then for every $\varepsilon > 0$, there exists a constant C dependend on the algebraic number a and ε such that for all positive integers p and q:

$$|a - \frac{p}{q}| > \frac{C}{q^{2+\varepsilon}}. \tag{1}$$

See [Roth(1955)].

With the help of the inequality for regular continued fractions from this follows

$b_{n+1} <= \frac{q_n^{\varepsilon}}{C}$

3 Formula by Bombieri and Van der Poorten

The main tool for the following proofs is the explicit formula for the coefficients of the regular continued fractions of algebraic numbers by [p.151, Theorem 3, formula (13)]Poorten, in which$f(x)$ is the minimal polynomial of the algebraic number. The formula connects d_n and b_n so it can be used here with ABC and Roth's theorem, which have no direct connection to d_n. In the main term of the formula,

$\frac{f'(x)}{f(x)}$ leads to d_n in the denominator. This holds because $\frac{f'(x)}{f(x)}$ results in $\frac{f'(\frac{p_n}{q_n})}{f(\frac{p_n}{q_n})} = \frac{q_n^s s p_n^{s-1}}{d_n q_n^{s-1}}$ for the convergents with $f(x) = x^s - k$ being the minimal polynomial of the sth root of k.

For the 3rd degree the error term of the formula [p.151, Theorem 3, formula (13)]Poorten can be ignored due to the effective bound by Liouville's theorem.

4 ABC Conjecture

To formulate the ABC Conjecture we need the following definition:

For a positive integer a, $rad(a)$ is the product of the distinct prime factors of a.

[ABC Conjecture] For every positive real number ε, there exists a constant K_ε such that for all triples (a, b, c) of coprime positive integers, with $a + b = c$:

$$c < K_\varepsilon \cdot rad(abc)^{1+\varepsilon}.$$

Note that this version of the ABC Conjecture is not effective. This is so because only the existence of a constant K_ε is claimed without specifying how to calculate K_ε for given ϵ.

5 Resulting ABC equations

We look at 3rd roots of k with k not a cubic number. The coefficients of the regular continued fraction are called b_n and the approximants $\frac{p_n}{q_n}$.

We call the equations with positive numbers

$p_n^3 = kq_n^3 + d_n$

resp

$kq_n^3 = p_n^3 + d_n$

resulting equations.

To use ABC we have to get rid off the cofactors of the resulting equations. For this purpose we observe as $\gcd(q_n, p_n) = 1$ that the gcd of p_n and k and the gcd of p_n and d_n are the same by divisionary considerations and so both are smaller or equal to k.

If the left side of

$p_n^3 = kq_n^3 + d_n$

is divisible by a factor than either k is also divisible (as the gcd of p_n and $q_n = 1$) and then d_n must be as well as the right side must be divisible by the factor or k is not divisible than d_n also can not be divisible as otherwise there would be a contradiction. And so both gcds must be the same.

These equal gcds will be called gcd in the following. It is larger or equal to 1 and smaller or equal to k.

So with
$\frac{p_n^3}{gcd]} = \frac{kq_n^3}{gcd} + \frac{d_n}{gcd}$
we get an ABC equation.

6 Construction of the bound for Roth's constant

ε is the ABC ε and K_ε the ABC constant.

ABC implies
$\frac{p_n^3}{gcd]} <= K_\varepsilon Rad(p_n q_n d_n k gcd^{-3})^{1+\varepsilon}$
This impiles
$p_n^3 <= K_\varepsilon (p_n q_n d_n k)^{1+\varepsilon} \frac{gcd}{gcd^{3+3\varepsilon}}$

For 3rd roots the formula [Bombieri & Van der Poorten(1975), p.151, Theorem 3, formula (13)] by Bombieri and Van der Poorten is valid without error term due to Liouville's theorem as given by Bombieri and Van der Poorten and their method can be used directly. For $n >= 4$ an error term can arise and the method can not be started with the first step. This is the reason, why we restrict ourselves to 3rd roots here.

From [Bombieri & Van der Poorten(1975), p.151, Theorem 3, formula (13)], we have the inequality
$d_n <= \frac{3p_n^2}{q_n b_{n+1}}$
Using that inequality we get
$b_{n+1} <= K_\varepsilon^{1/(1+\varepsilon)} 3kp_n^{\frac{3\varepsilon}{1+\varepsilon}} \frac{gcd}{gcd^{3+3\varepsilon}}^{1/(1+\varepsilon)}$
Roth's theorem implies the existence of a constant C depended only on ε_{Roth} and 3rd root of k with
$b_{n+1} <= \frac{q_n^{\varepsilon_{Roth}}}{C}$
So as relationship between the epsilons we get
$\varepsilon_{Roth} = 3\frac{\varepsilon_{ABC}}{1+\varepsilon_{ABC}}$
and we get that the inverse of Roth's constant C can be bounded by
$K_\varepsilon^{\frac{1}{1+\varepsilon}} 3k\frac{p_n}{q_n}^{\frac{3\varepsilon}{1+\varepsilon}} \frac{gcd}{gcd^{3+3\varepsilon}}^{1/(1+\varepsilon)}$
This is clearly possible as $\frac{p_n}{q_n}$ converges to 3rd root of k and the gcd is bounded by 1 and k.

7 Range of the Epsilons

For $\varepsilon_{ABC} = \varepsilon_{Roth} = 0$ it results that b_n is bounded. This is an open question, but it is known that ABC is not valid for $\varepsilon = 0$, but then known counterexamples are not resulting equations of 3rd roots.

For $\varepsilon_{ABC} = \varepsilon = \frac{1}{2}$ we get $b_{n+1} <= q_n \frac{p_n}{q_n} factor$, which bascially is Liouville's theorem. So the interesting part is ε_{ABC} between 0 and $\frac{1}{2}$.

8 Example $\sqrt[3]{2}$

The regular continued fraction starts with $[1; 3, 1, 5, ..]$

The first resulting equations are

2 = 1 + 1, which is an ABC equation

64 = 54 + 10, where $gcd = 2$ so we get the ABC equation $32 = 27 + 5$

128 = 125 + 3, which is an ABC equation

Our formula for a bound of the inverse of Roth's constant

$K_\varepsilon^{\frac{1}{1+\varepsilon}} 3k \frac{p_n}{q_n}^{\frac{3\varepsilon}{1+\varepsilon}} \frac{gcd}{gcd^{3+3\varepsilon}}^{1/(1+\varepsilon)}$

leads in this case to

$K_\varepsilon^{\frac{1}{1+\varepsilon}} 6 \frac{4}{3}^{\frac{3\varepsilon}{1+\varepsilon}}$

as $\frac{4}{3}$ is the largest approximant and the factor with $gcd = 1$ is larger than the one with $gcd = 2$, so that it can be taken and is equal to 1.

There is a lot of data on ABC, which could be used to get a bound for the inverse of Roth's constant within the range already analysed by computer calculation. We will investigate $\sqrt[3]{2}$ now for 2 different ε.

8.1 $\varepsilon = 0.5$

Due to Korobov's famous result [Korobov(1990)]

$$| \sqrt[3]{2} - \frac{p}{q} |> \frac{1}{q^{2.5}}$$

for all natural numbers p and q with the exception of 1 and 4 for q

If we ignore 1 then Roth's constant is calculated by

$$| \sqrt[3]{2} - \frac{5}{4} | 4^{2.5}$$

which is approximately 0,32 and so the the inverse is given by

$\frac{1}{C} = 3,125$ appreximately.

Now we use our way from the ABC side. A $\varepsilon_{Roth} = 0.5$ leads to an $\varepsilon_{ABC} = \frac{1}{5}$

Again ignoring the first case 2 = 1 + 1 from the ABC data it can be calculated within the range analysed by computer calculation that the ABC K_ε results from the resulting equation 128 = 125 + 3.

So calculation gives $K_{\frac{1}{5}}$ is approximately 2,16

So our formula for a bound of the inverse of Roth's constant results in

$\frac{1}{C} <= 2,16^{\frac{5}{6}} 6 \frac{4}{3}^{\frac{1}{2}}$

which is approximately 13,16

So in this case the road via ABC leads to a loss in precision, which is mainly due to the use of $\frac{4}{3}$ as upper bound.

8.2 $\varepsilon = 0.4$

Here the result is not known exactly. In the proof of corrollary 2.2 [Voutier(2007)] gives one result using the hypergeometric method for $\varepsilon = 0.4325$ using his theorem 2.1 (p.285):

$$| \sqrt[3]{2} - \frac{p}{q} |> \frac{10^{-99}}{q^{2.4321}}$$

So the inverse of Roth's constant is bounded by 10^{99}.

Now we use our way from the ABC side and use a $\varepsilon_{Roth} = 0.4$. This leads to an $\varepsilon_{ABC} = \frac{2}{13}$

Again ignoring the first case 2 = 1 + 1 from the ABC data it can be calculated within the range analysed by computer calculation that the ABC K_ε results from the resulting equation 128 = 125 + 3.

So calculation gives $K_{\frac{2}{13}}$ is approximately 2,527

So our formula for a bound of the inverse of Roth's constant results in

$\frac{1}{C} <= 2,527^{\frac{13}{15}} 6^{\frac{4}{3}}{}^{\frac{6}{15}}$

which is approximately 15,03

8.3 Table for bounds of the inverse of Roth's constant C

We conclude with a table. Again we ignore the resulting equation 2 = 1 + 1 and assume that the resulting K_ε comes from the resulting equation 128 = 125 + 3. This is not always the case as will become clear from the data. Only with ε large enough in the known data range it comes from that equation.

ε_{Roth}	ε_{ABC}	K_ε	ABC bound for $\frac{1}{C}$	Known bound for $\frac{1}{C}$
0	0	4,267	60,686	Probably unbounded
0,4	$\frac{2}{13}$	2,5	15,03	Not known
0,5	$\frac{1}{5}$	2,161	13,16	3,125 (Korobov is exact)
1	$\frac{1}{2}$	0,78	6,78	6 (Liouville can be lowered to 1,575)

Table 1: Approximate values for $\sqrt[3]{2}$ as long as this resulting K_ε can be taken

So the resulting K_ε for $\varepsilon = 0$ only comes from 128 = 125 + 3 as long as the b_n are bounded by 60. This is the case until $b_{36} = 534$ is larger. In general the bound resulting from this ABC method is very smooth, but not as good as the exact results of course. Of course if ABC is proved for an explicit ε then that K_ε can be taken and no numerical data would be needed. At the moment one open conjecture is that ABC holds with $\varepsilon = 1$ and $K_\varepsilon = 1$. But to get results for Roth's constant C a much smaller ε would be needed.

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